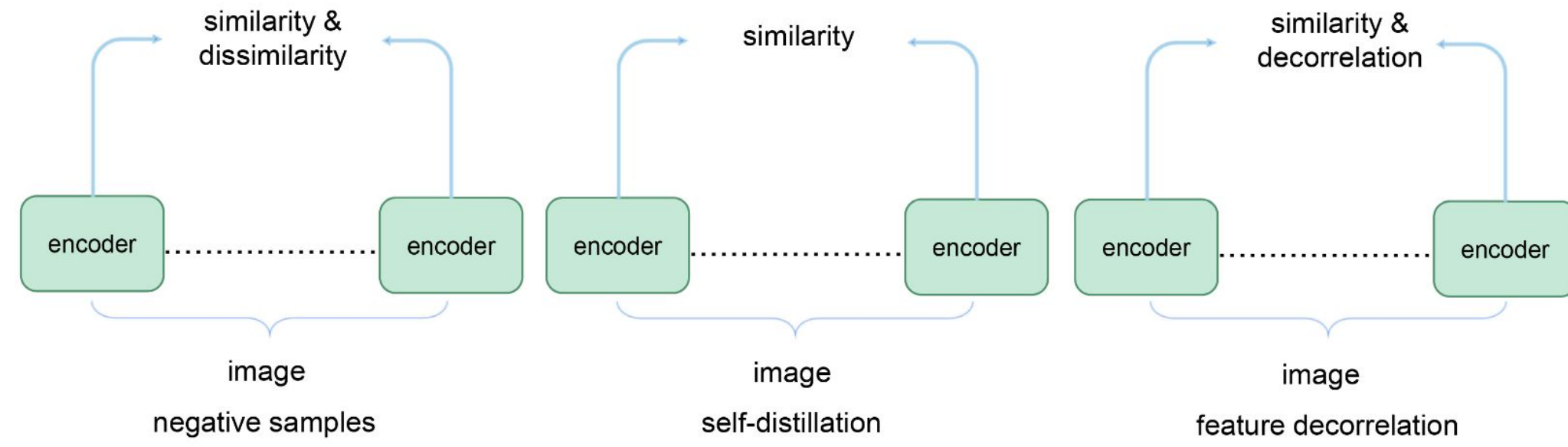


Definitions

Self Supervised Learning (SSL) - Generates labels by augmenting the input data. A specific loss function is used to learn representations that are invariant to augmentations made on the input data.

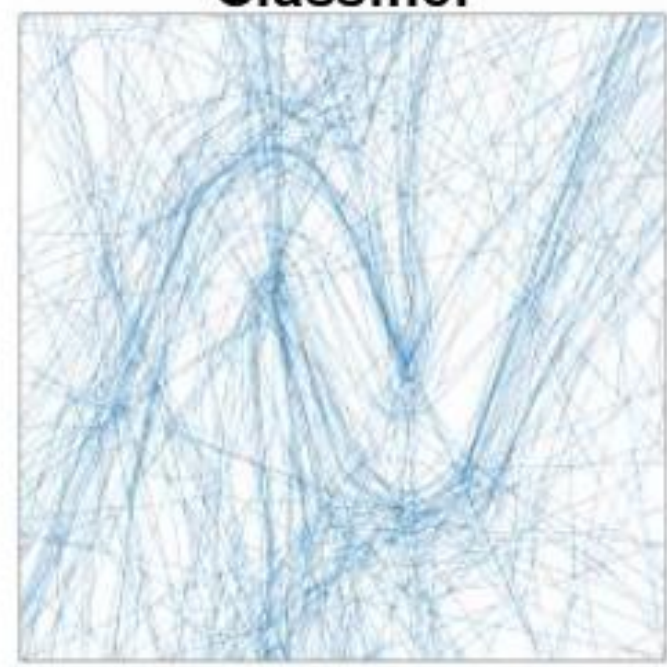
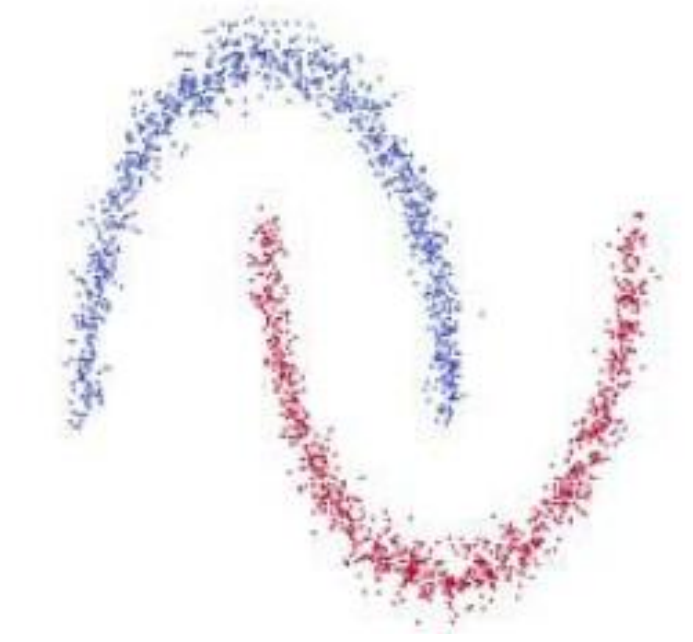


Taken from Gui, J., Chen, T., Zhong, J., Cao, Q., Sun, Z., Luo, H., & Tao, D. (2024). A survey on self-supervised learning: Algorithms, applications, and future trends. IEEE Transactions on Pattern Analysis and Machine Intelligence, 46(12)

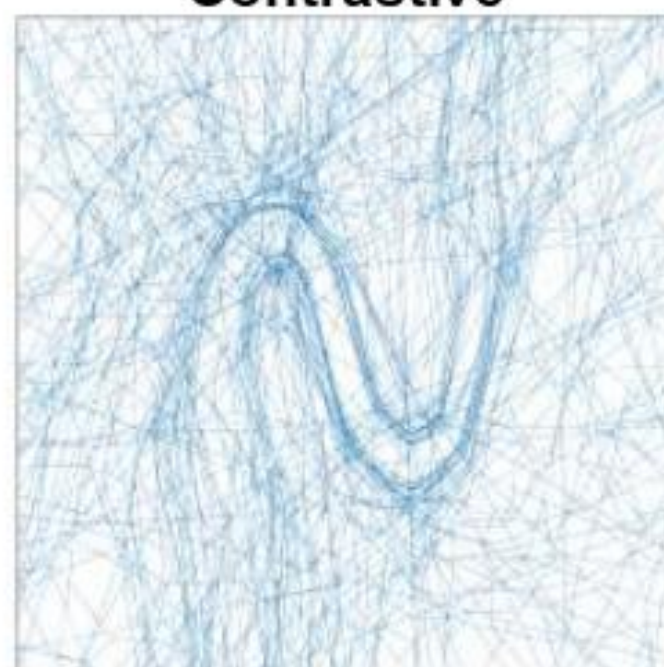
Linear regions - A neural networks piecewise linear activation function (ReLU) partitions the input space into regions (polytopes).

$$\sigma(z) = \max(0, z)$$

Supervised
Classifier

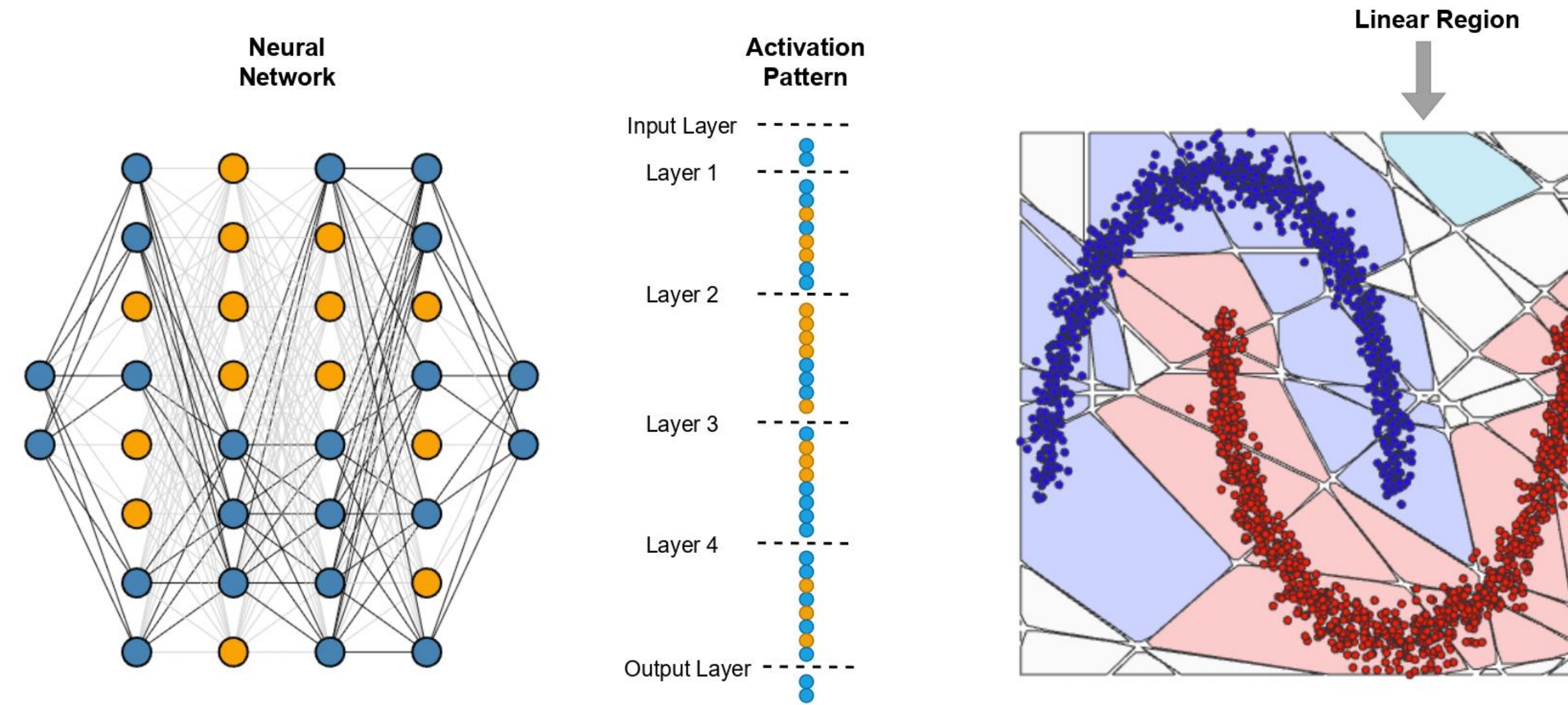


Supervised
Contrastive



Problem Statement

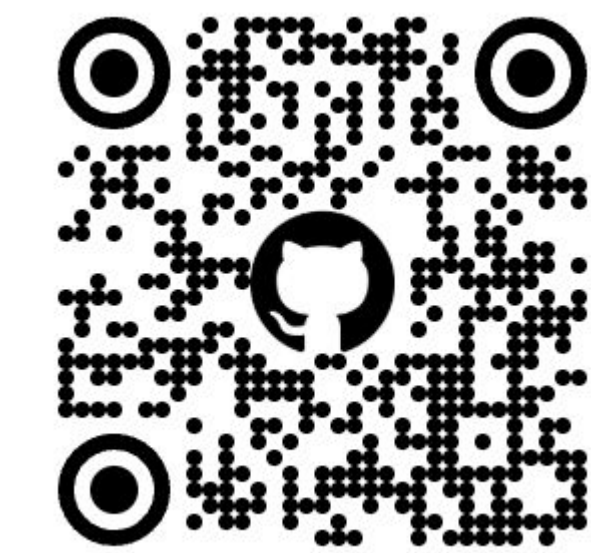
- Understanding the structure and distribution of regions near the data distribution has become an active area of research [1, 2].
- Practitioners count the number of regions as a measure of network complexity.
- Most current work focuses on analysing the linear regions generated by supervised classifier networks [3, 4].
- However, the geometric properties of Self-Supervised Learning (SSL) representations remain largely underexplored.



Motivation

- Linear regions (polytopes) are considered the atomic unit of neural networks.
- Any inputs that belong to the same polytope are subject to the same affine transformation.
- Nearby polytopes implement similar transformations, hence we can consider neighbouring regions as the molecules of a network.
- Assessing the quality and evolution of linear regions provides an interesting framework for preserving the structure of the embedding space.

website

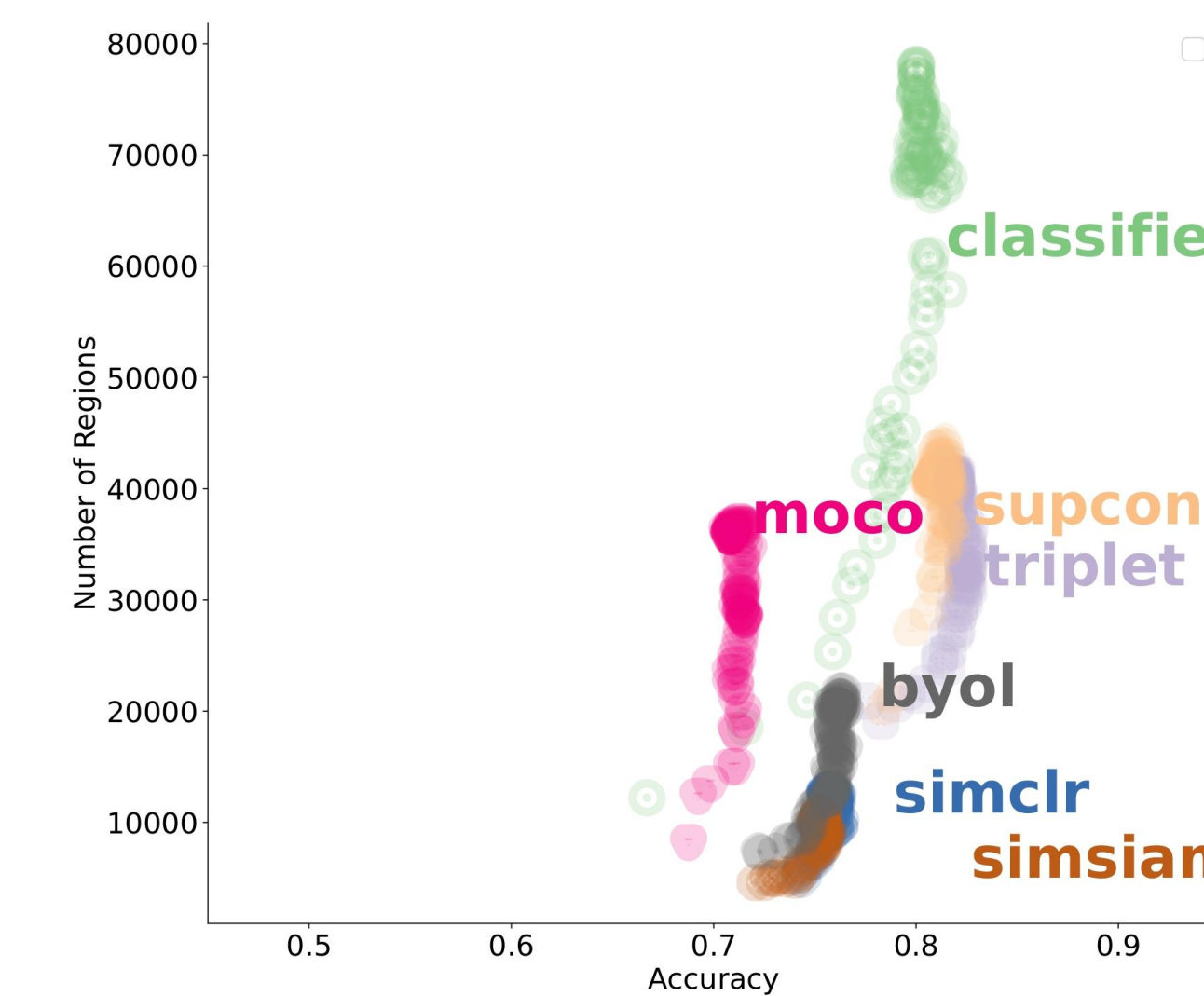


code

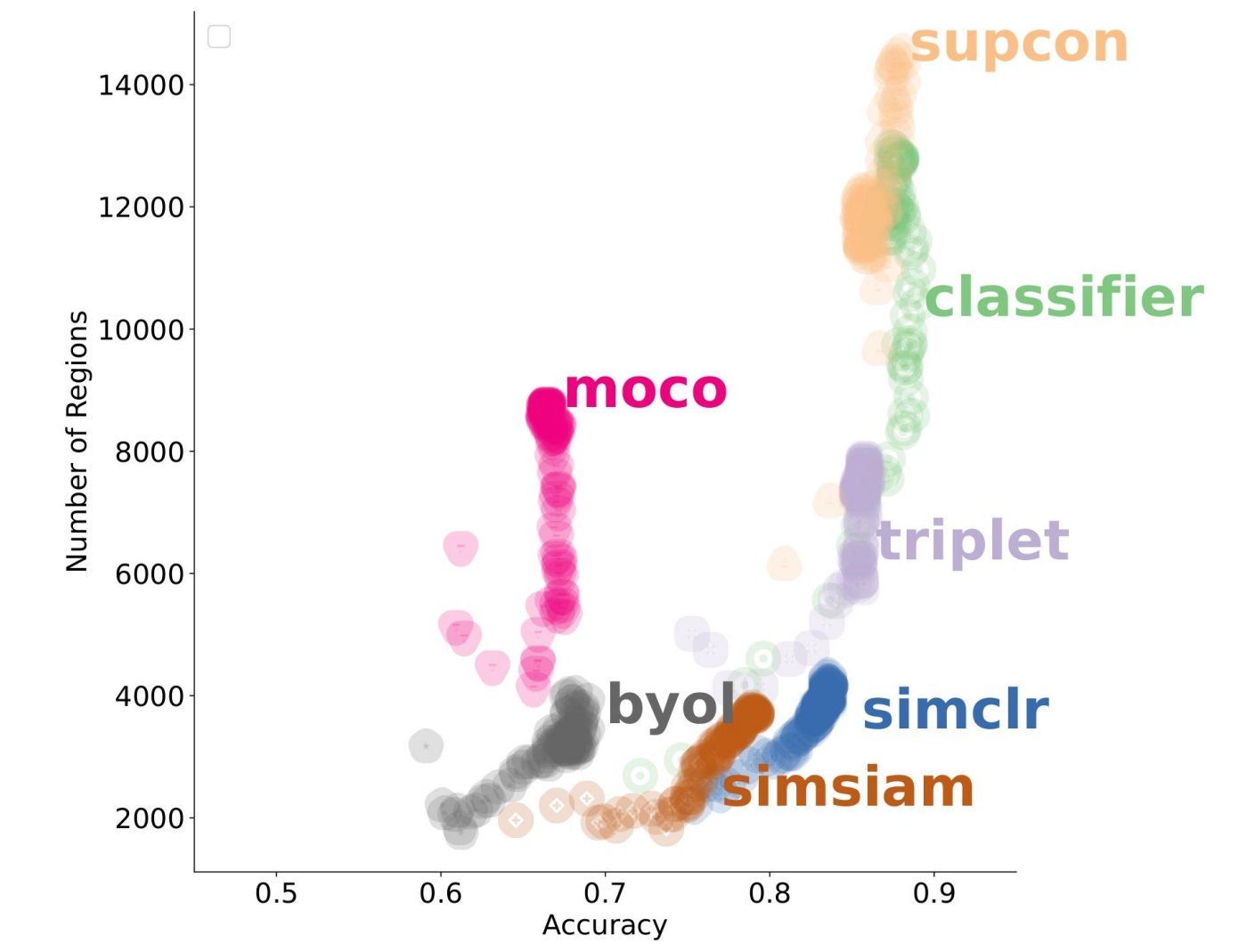


Results

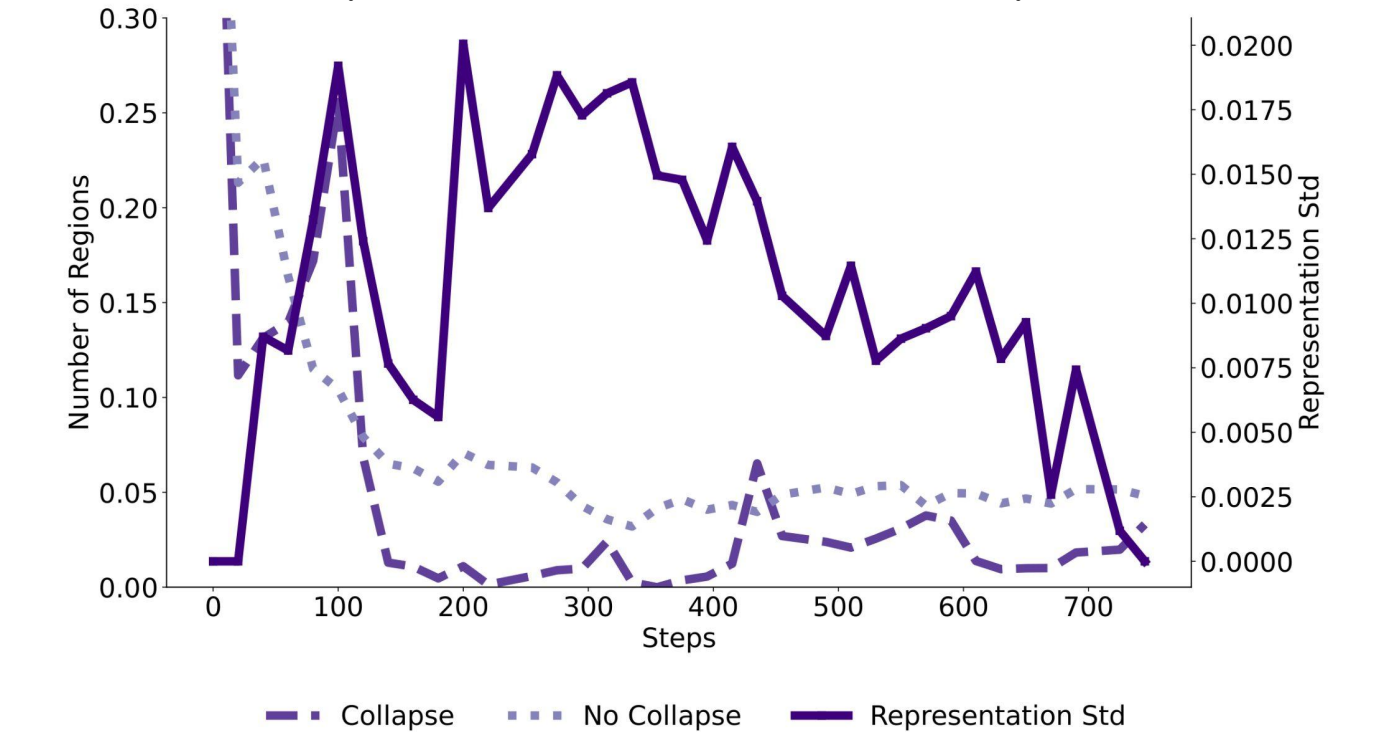
FashionMNIST



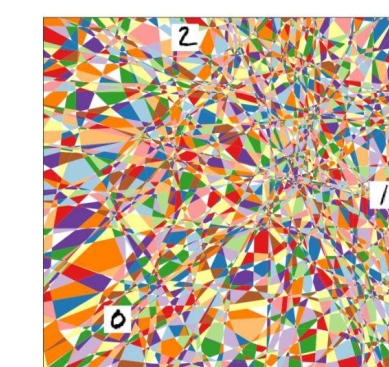
MNIST



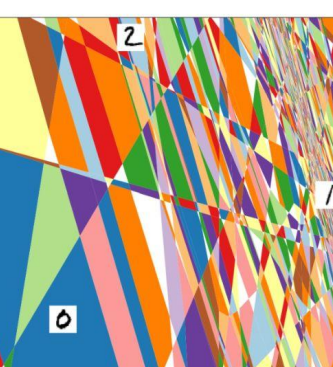
Representation Collapse



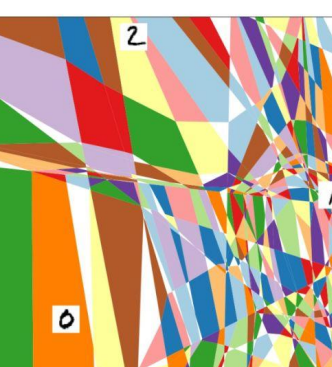
Initialization



Step 100

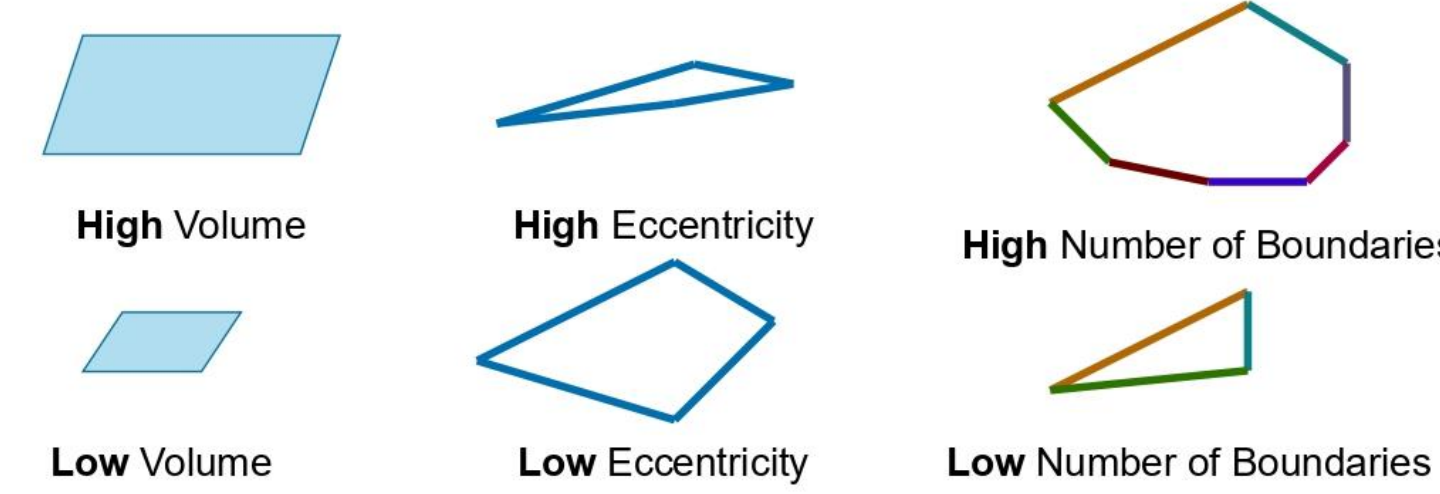


Collapse



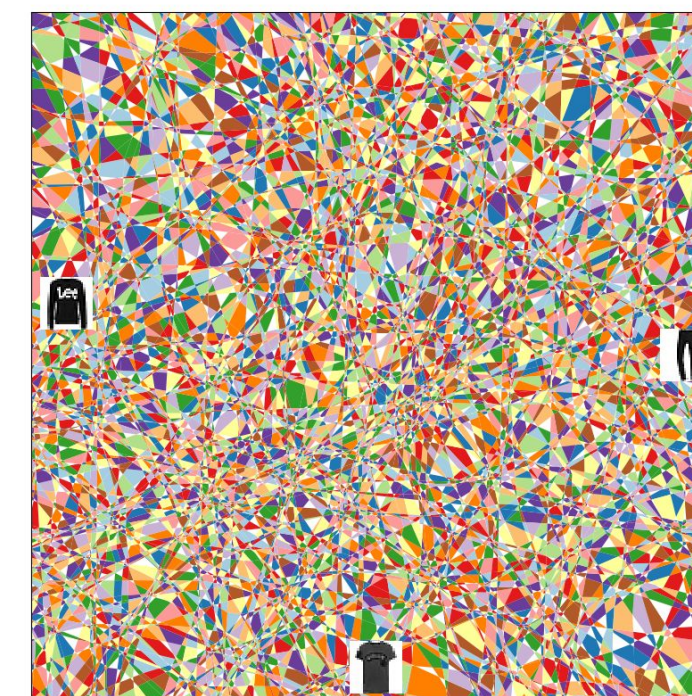
Methodology

- Transform a high-dimensional input space into a two-dimensional space and partition it to obtain its linear regions [4].
- Adopt *SplineCam* for interpretation and visualisation of the exact number of regions from a two-dimensional input space [5].



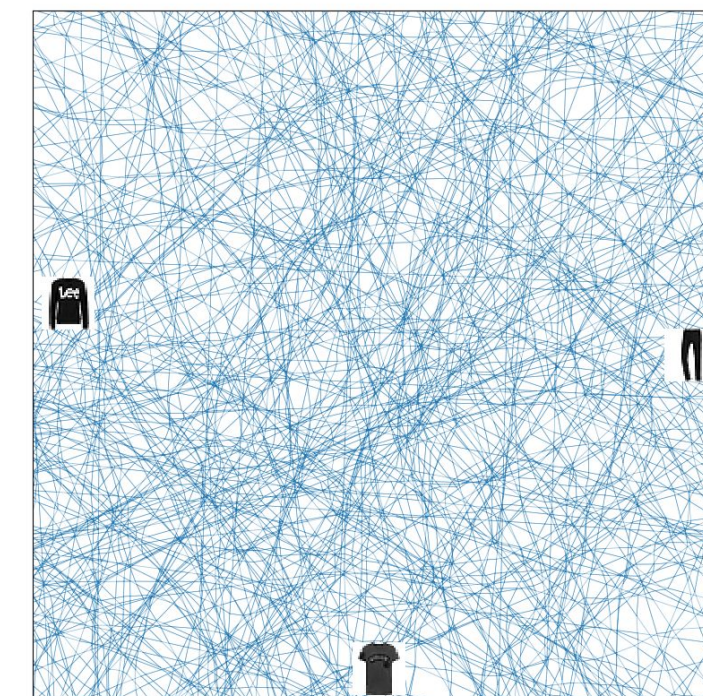
input space

$$\mathbb{R}^d \approx \bigcup_{i=1}^{N_{\mathcal{R}}} \mathcal{R}_i$$



volume

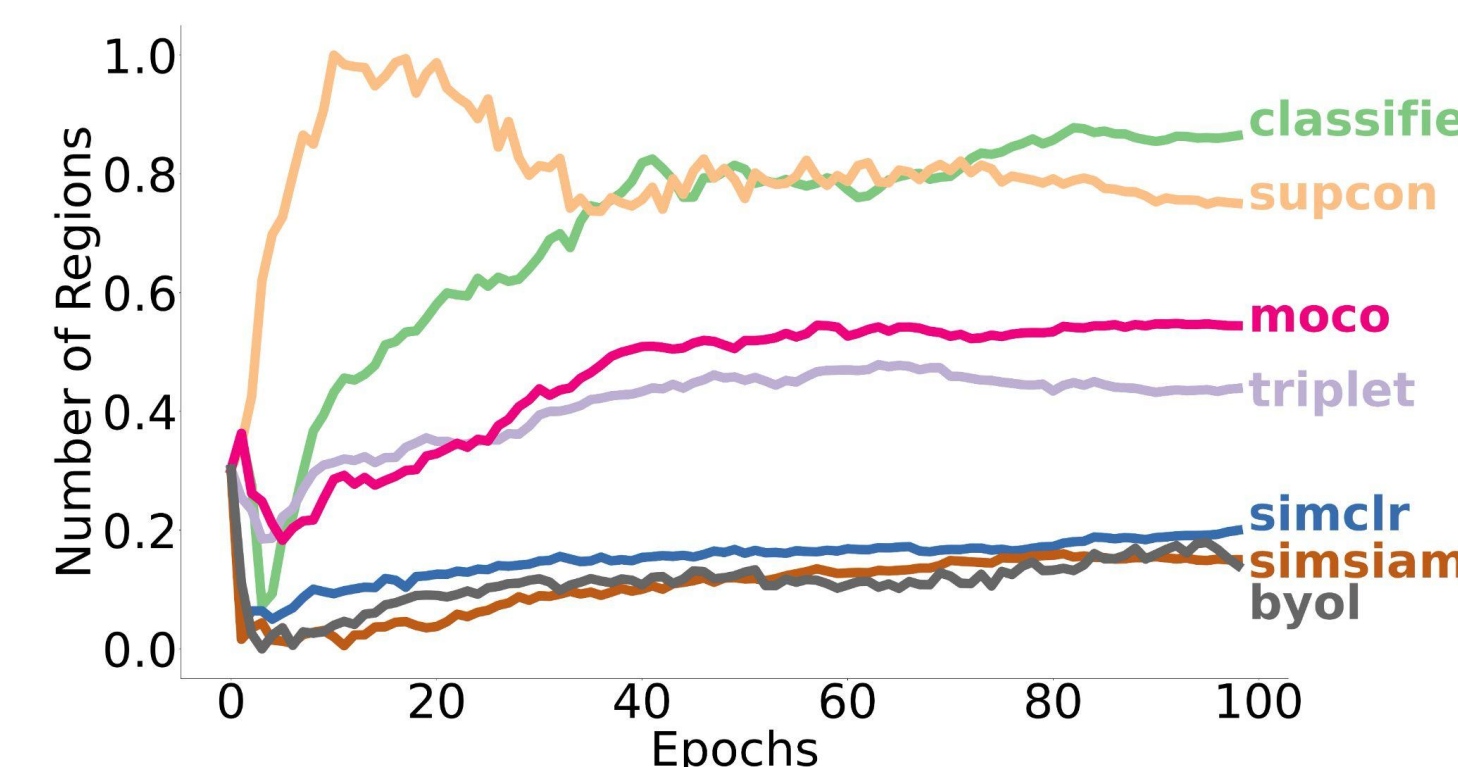
$$V_i = \frac{1}{2} \left| \sum_{k=0}^{n-1} (a_k b_{k+1} - b_k a_{k+1}) \right|$$



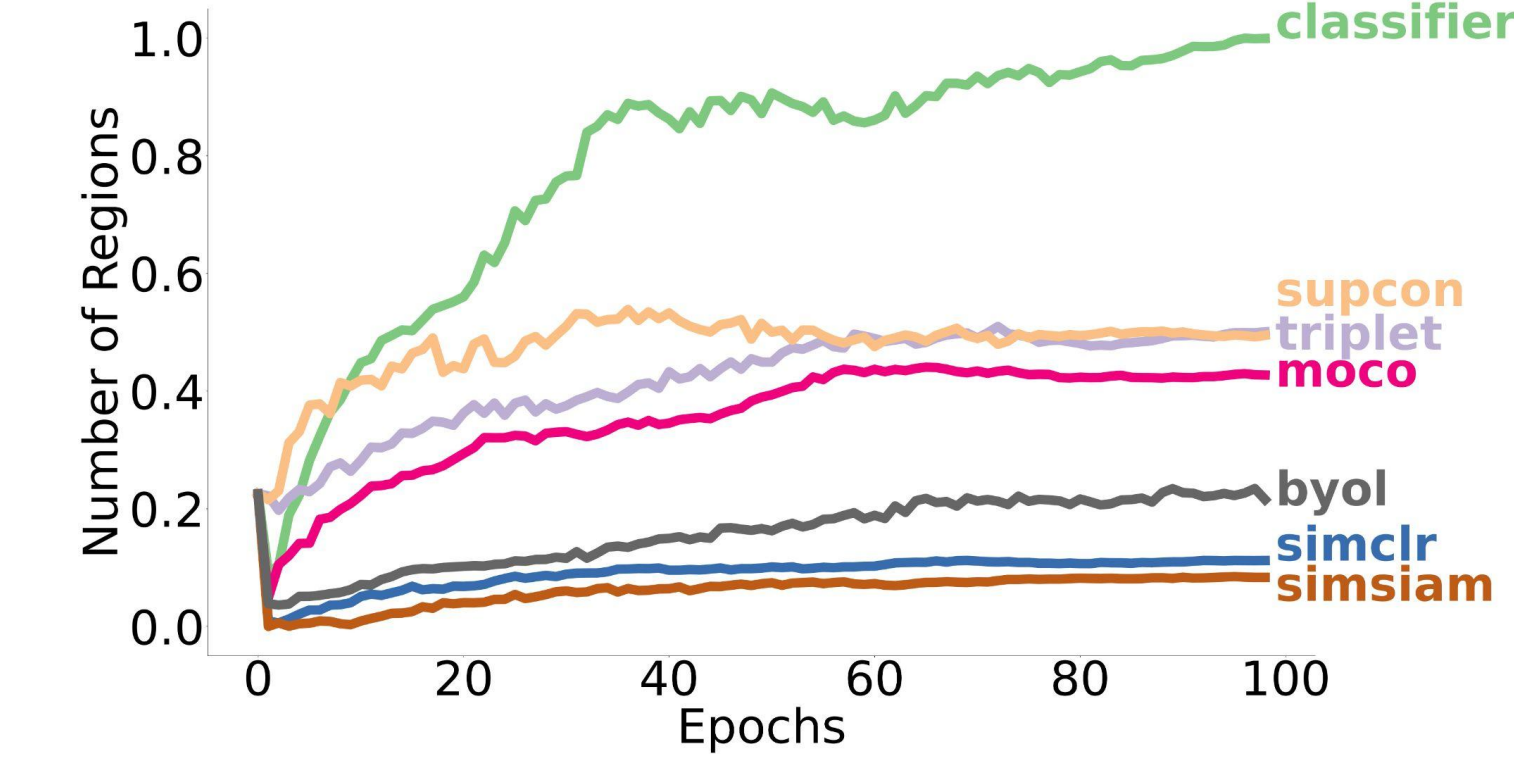
Eccentricity

$$E_i = \sqrt{1 - \left(\frac{\min(\Delta_a, \Delta_b)}{\max(\Delta_a, \Delta_b)} \right)^2}$$

MNIST



FashionMNIST



Future Work

- Future research could investigate techniques to directly optimise the polytopal space
- Discover methods that encourage fewer regions with greater volume and controlled eccentricity to improve the robustness and transferability of the model.
- The geometry of the polytopal space could provide an early indicator of representation collapse.
- In continual learning, researchers could leverage self-distillation methods to create as few regions as possible while avoiding collapse within the current task.

References

- Guido Mont'ufar, Razvan Pascanu, Kyunghyun Cho, and Yoshua Bengio. On the number of linear regions of deep neural networks. Advances in neural information processing systems, 27, 2014.
- David Rolnick and Max Tegmark. The power of deeper networks for expressing natural functions. arXiv preprint arXiv:1705.05502, 2017.
- Boris Hanin and David Rolnick. Complexity of linear regions in deep networks. In International Conference on Machine Learning, pages 2596-2604. PMLR, 2019.
- hmed Imtiaz Humayun, Randall Balestriero, and Richard Baraniuk. Deep networks always grok and here is why. arXiv preprint arXiv:2402.15555, 2024.
- hmed Imtiaz Humayun, Randall Balestriero, Guha Balakrishnan, and Richard G Baraniuk. Splinecam: Exact visualization and characterization of deep network geometry and decision boundaries. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 3789-3798, 2023.

M. Muthivhi



TL. van Zyl

